Joseph Naranjo

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About Me

Electronic Engineer specializing in AI architectures, real-time sensor data pipelines, and hardware-software integration. Experienced in designing multi-sensor data acquisition systems, electrical control infrastructure, and deploying machine learning models within industrial environments. Proven ability to act as a technical advisor, bridging complex physical systems with scalable backend infrastructures to navigate high-ambiguity engineering challenges.

Key Achievements

- **NASA Space Apps Challenge**: Global Finalist — Top 45 Worldwide (2025)

Experience

- **Micro1** — Palo Alto, CA (Remote)
  AI Training Specialist (Electronics & EDA) — *Apr 2026 – Present*
  - **Expert Verification & Advisory**: Act as a Subject Matter Expert (SME) to evaluate complex engineering scenarios and validate AI-generated electronic designs, ensuring adherence to strict electrical engineering principles.
  - **Circuit Simulation & Failure Modes**: Develop and execute advanced simulation test cases using **ngspice** and **Qucs-S** to benchmark theoretical accuracy, verify SPICE netlists, and identify failure modes under incomplete data conditions.
  - **Technical Scenario Design**: Design high-complexity sample tasks and optimized context logic to guide Large Language Models (LLMs) in solving novel, high-stakes hardware-software integration challenges.
  - **Trade-off Assessment**: Analyze and document critical trade-offs between system cost, safety, constraints, and performance to inform pipeline development and research directions.
- **Orillia WoodWorking Inc.** — Orillia, ON
  Automation & Systems Engineer — *Full-Time · Aug 2024 – Present*
  - **Industrial Maintenance & MTTR Optimization**: Performed preventive and corrective maintenance on motors, VFDs, and control systems; **reduced machine downtime by 10%** and improved **Mean Time To Repair (MTTR) by 50%** by implementing a historical fault-tracking system for faster diagnostics.
  - **IoT & Data Acquisition Architecture**: Designed and deployed an IoT-integrated electrical panel with PLCs and Node-RED, enabling real-time telemetry and redundant storage (PostgreSQL/Firebase) for **100% data traceability** of critical machine assets.
  - **ERP & Administrative Efficiency**: Architected a custom PostgreSQL-based ERP, **reducing administrative processing time by 40%** through automated material tracking, sales orders, and inventory synchronization for custom manufacturing workflows.
  - **Operational Accuracy**: Eliminated inventory inconsistencies and minimized manual entry errors by developing a unified CMMS/ERP platform, providing a **single source of truth** for production and maintenance KPIs.
  - **Canadian Standards Compliance**: Ensured all electrical and software implementations met Ontario industrial safety standards while collaborating with cross-functional teams to optimize plant-floor operations.
- **NDigital Horizon** — Barrie, ON
  Founder & Lead AI Solutions Architect — *Self-Employed · Jan 2024 – Present*
  - **Technology Consultancy**: Founded and operate a technology consultancy focused on AI automation and IIoT solutions for SMBs across Canada and Latin America.
  - **Audi-Conta**: Architected a tax automation platform that processes SRI XML invoices using AI (Vertex AI + Claude), reducing administrative processing time for SMB clients.
  - **DK Sport Arena**: Developed a booking and operations management system for DK Sport Arena, automating scheduling and client workflows.
  - **Loxone Official Partner**: Official Loxone Partner, certified in professional building automation and smart home systems. Currently offering home and commercial automation services through NDigital Horizon's dedicated automation division (automation.ndigitalhorizon.com).

Technical Skills

- **Engineering & Automation**: Industrial Automation, Electrical Panel Design, PLCs, VFDs, Contactors, Motor Control, Modbus Communication, Predictive Maintenance, IIoT, EDA Tools (**ngspice**, **Qucs-S**)
- **Programming & Systems**: Python, SQL, Node-RED, Thingsboard, PostgreSQL, Firebase, Git, REST APIs, Matlab, PyTorch, scikit-learn
- **AI & Data Systems**: Real-Time Data Processing, Sensor Data Ingestion, Multi-Sensor Synchronization, Data Pipelines, Anomaly Detection, Prompt Engineering
- **Infrastructure**: Docker, Cloud Services (AWS, Google), Industrial Networking

Publications

- **Research Paper**: Naranjo, J. (Author). "Application of Neural Networks and Computer Vision for NFT Hydroponic Crop Control Systems." Published in Revista Social Fronteriza.

Education

- **Georgian College** Barrie, ON, CA
  Graduate Certificate - Artificial Intelligence: Architecture, Design, and Implementation
- **Salesian Polytechnic University** Quito, EC
  Bachelor of Engineering - Electronic Engineering

Projects

- **Industrial Data Acquisition & Monitoring System — IIoT Architecture**: Designed and deployed a production-ready real-time monitoring system for a manufacturing environment. Integrated industrial-grade hardware (PLCs, gateways) with a multi-layer data backend, enabling continuous sensor data ingestion, KPI computation, and operational visibility across the plant floor.

  Tech: Node-RED · PostgreSQL · Firebase · Docker · Industrial IoT · REST APIs
- **Smart Diagnostics Failure — AI-Powered Anomaly Detection System**: Developed an AI-powered system for industrial equipment failure detection using raw sensor data. Built and trained machine learning models in Python and PyTorch for anomaly detection and predictive analysis. Implemented a Modbus-based data acquisition layer and designed a scalable processing pipeline architected for both edge and cloud deployment.

  Tech: Python · PyTorch · Modbus · scikit-learn · Real-Time Data Pipeline

Certifications

- **Certifications**: AWS Academy Machine Learning Foundations (2025)
  Google Developers Event — Participation Certificate (Toronto, 2025)